

Project Oracle: AI-Powered Trading Intelligence

Elevating the TradingView experience from visualization to systematic cognition.

PRODUCT NAME	Project Oracle	STATUS	Draft / Proposal
PRODUCT MANAGER	[AI Innovation Team]	TARGET DATE	Q3 - Q4 2026
COMPANY	Dante Labs	DOC ID	DANTE-PRD-001

1. Executive Summary

Project Oracle aims to solve the core problem of "analysis paralysis" in retail trading. By integrating a multi-modal AI layer into TradingView, we empower users to generate institutional-grade strategies using natural language, detect complex patterns via Computer Vision, and understand market sentiment in real-time. This PRD outlines the functional vision required to outperform commoditized charting competitors.

2. Target Audience

- **The "Time-Poor" Professional:** Wants complex strategies set up quickly without learning Pine Script.
- **The Strategic Learner:** Uses AI to understand why a certain pattern is forming.
- **The Systematic Edge-Seeker:** Needs real-time sentiment data to confirm technical signals.

3. User Stories

As a Trader, I want to describe my entry rules in plain English **so that** the system can instantly generate a backtested Pine Script without me having to write code.

As a Newbie, I want the AI to "label" the chart patterns (like Head & Shoulders) visually **so that** I can learn to identify setups with high probability.

As a Power User, I want to see a "Sentiment Heatmap" overlaid on my candles **so that** I can avoid trading into news-driven volatility spikes.

4. Functional Requirements

ID	Requirement Name	Priority	Description
FR-01	AI Side-Panel UI	P0	A React-based drawer inside the TradingView frame for AI interaction.
FR-02	Natural Language to Pine	P0	LLM integration that outputs valid, backtestable Pine Script v5.
FR-03	Visual Pattern Labeling	P1	Computer Vision model to draw pattern boundaries on the active chart.
FR-04	Contextual Sentiment Index	P2	Real-time score (Bearish/Bullish) based on X, Discord, and News data.
FR-05	"One-Click" Deploy	P1	Button to push generated strategies directly to the broker connection.

5. Success Metrics (KPIs)

- **MAU Growth:** 25% increase in Monthly Active Users within 3 months of launch.
- **Engagement:** Average time spent on chart per session increased by 40%.
- **Scripting Efficiency:** Reduction in manual "User Script Errors" support tickets by 60%.
- **Monetization:** 15% conversion rate to the "Premium AI" subscription tier.

6. Non-Functional Requirements

- **Latency:** Side-panel responsiveness must be < 200ms.
- **Accuracy:** AI pattern detection must have a verified False Positive Rate < 15%.
- **Device Support:** Must function on Desktop and Mobile via the Dante Labs web app.

7. Risks & Mitigations

Risk	Impact	Mitigation Strategy
Hallucinated Scripts	High	Implement a Pine Script linter/validator before presenting code to user.
High Inference Cost	Med	Utilize model quantization and implement request rate-limiting.

Data Latency	High	Use direct WebSocket feeds for raw data; prioritize local inference for pattern vision.
--------------	------	---